

Reviewer Briefing Note

Project: Amnesty Human Rights DAO (HRDAO)

Milestone 2: Tokenomics Modelling & Simulation

Project ID: 1300018

■ Purpose of Milestone 2

Deliver a complete, audited, and reproducible **tokenomics and smart-contract framework** for the Human Rights DAO.

This milestone demonstrates the feasibility, security, and sustainability of a dual-token governance system and establishes readiness for Milestone 3 (MVP prototype).

■ Key Achievements in this Deliverable

1. Comprehensive Tokenomics Model

- Dual-token design: **HRT-UTL (Utility)** and **HRT-GOV (Governance Soulbound)**.
- Adaptive monetary policy (2–8 % inflation band + deflationary sinks, vesting architecture).
- Rewards mapped to verifiable Amnesty actions (petitions, education, protests, human-energy inputs).
- Anti-gaming and Sybil-resistance logic tested and documented.

2. Simulation & Methodology

- Two 12-month simulation runs (baseline and crisis year) using real Amnesty data.
- Outputs confirm stable token velocity (4–8 turns p.a.) and inflation control within target band.
- Full parameter sheets, CSV datasets, and charts published on public GitHub.

3. Governance Architecture

- Quadratic voting, delegation (DReps), and proof-of-liveness mechanisms.
- CIP-68 Soulbound governance token anchors identity and reputation.
- “Embodied Governance” mapping links human-rights principles to on-chain rules (e.g., collective stewardship → multi-sig treasury).

4. Dual Independent Audits Completed

- **Technical Audit 1 – Dr Sagar Dake (PhD, AUT NZ):**
 - 12 findings reviewed; no critical issues.
 - Serialization and nonce validation resolved; multisig mint and commit–reveal pattern scheduled for M3.
 - Confirmed contracts fit for pilot deployment and aligned with Aiken/Plutus V2 best practice.

- **Technical Audit 2 – Chereese Eriepa (Software Engineer / Ops Lead):**
 - All unit tests passed; no security breaches or leakage.
 - Verified treasury/oracle signatory logic and vault authenticity.
 - Execution units well within protocol limits (≤ 135 K mem / 51 M CPU).
 - Only minor notes on comment clarity and edge-case tests — both addressed in latest commit.

5. Scalability & Verification Path

- Architecture pre-configured for Hydra heads and Mithril snapshots for parallel processing.
- Layered identity approach (self-attest → peer → oracle → zk-ID future module).

6. Delivery Evidence & Transparency

- Acceptance criteria mapped line-by-line to outputs in the *Milestone 2 Execution & Alignment* framework.
- GitHub repository includes simulation models, contract stubs, audit reports, and README summary.
- Visual charts and tables demonstrate allocation, supply dynamics, and governance flows.

■ Reviewer Takeaway

This milestone **fully satisfies and exceeds Catalyst Milestone 2 acceptance criteria** across accuracy, functionality, security, scalability, and transparency.

Both independent audits confirm:

- No critical or major vulnerabilities,
- Secure and efficient contract execution,
- Robust alignment between tokenomics design and smart-contract logic.

The deliverable is technically mature, openly documented, and mission-aligned — positioning Amnesty HRDAO as a flagship example of **decentralised governance for social impact** within the Cardano ecosystem and ready for Milestone 3 (MVP deployment).